Land 2: Cellspire Grove

Chemistry and the cell · OpenStax Ch. 2 and 3

Key ideas

Atoms, ions, and bonds

- Atoms contain protons (positive) and neutrons (neutral) in the nucleus, with electrons (negative) in shells around it.
- Atomic number equals the number of protons; mass number equals protons plus neutrons.
- Isotopes have the same number of protons but different numbers of neutrons; radioisotopes emit radiation and are used in PET scans and cancer treatment.
- An atom that loses electrons becomes a positive cation (sodium ion, potassium ion); one that gains electrons becomes a negative anion (chloride ion).
- Ionic bonds are attractions between oppositely charged ions; covalent bonds share electrons, equally (nonpolar) or unequally (polar, as within water).
- Hydrogen bonds are weak attractions between polar molecules; they hold the two DNA strands together and help shape proteins.

Water, reactions, pH, and buffers

- Water is polar and an excellent solvent, with a high heat capacity and a high heat of vaporization, so evaporating sweat cools the body.
- Synthesis builds larger molecules (anabolic); decomposition breaks them apart (catabolic); exchange reactions do both.
- Dehydration synthesis joins monomers and releases water; hydrolysis adds water to split polymers.
- pH runs from 0 to 14; below 7 is acidic, 7 is neutral, above 7 is basic; each unit is a tenfold change in hydrogen ion concentration.
- Normal blood pH is 7.35 to 7.45; below that is acidosis, above it is alkalosis.
- Buffers resist pH change; the bicarbonate system pairs carbonic acid with bicarbonate, and losing carbon dioxide raises pH.

Carbohydrates, lipids, and proteins

- Monosaccharides: glucose, fructose, galactose; disaccharides: sucrose (glucose plus fructose), lactose, maltose; polysaccharides: glycogen (animal storage), starch, cellulose.
- Triglycerides are glycerol plus three fatty acids and store energy; saturated fats have no double bonds and are solid at room temperature.
- Phospholipids have a hydrophilic phosphate head and two hydrophobic tails and form membranes.
- Steroids have four fused rings; cholesterol is used to make steroid hormones such as estrogen and testosterone.
- Proteins are chains of amino acids joined by peptide bonds; structure levels are primary, secondary, tertiary, and quaternary.
- Heat or abnormal pH denatures proteins, unfolding their shape and destroying function.

Nucleic acids, ATP, and enzymes

- Nucleotides contain a five carbon sugar, a phosphate group, and a nitrogenous base.
- DNA is double stranded with deoxyribose and bases A, T, G, C (A pairs with T, G with C); RNA is single stranded with ribose and uracil in place of thymine.
- ATP is a nucleotide with three phosphates; hydrolysis of the last phosphate bond releases energy for cell work.
- Enzymes are mostly proteins that lower activation energy; the substrate binds the active site, and the enzyme is not used up.
- Enzyme activity depends on temperature and pH; inhibitors that block the active site slow the reaction.

Plasma membrane and passive transport

- Fluid mosaic model: phospholipid bilayer with embedded proteins, cholesterol that stabilizes fluidity, and a glycocalyx used for cell recognition.
- Simple diffusion moves small nonpolar molecules such as oxygen and carbon dioxide from high to low concentration without ATP.
- Facilitated diffusion uses channel or carrier proteins for ions and polar molecules such as glucose, still down the gradient and without ATP.
- Osmosis is water diffusing across a selectively permeable membrane toward higher solute concentration.
- Isotonic (0.9 percent saline): no net change; hypotonic: cells swell and may lyse; hypertonic: cells shrink (crenate).

Active and vesicular transport

- Active transport moves substances against their gradient using energy.
- The Na+/K+ pump uses one ATP to move 3 sodium ions out and 2 potassium ions in; this is primary active transport.
- Secondary active transport uses a gradient made by a pump, such as the sodium gradient, to move another substance.
- Endocytosis brings material in: phagocytosis (cell eating), pinocytosis (cell drinking), and receptor mediated endocytosis (specific, as for LDL cholesterol).
- Exocytosis releases vesicle contents, such as hormones, enzymes, and neurotransmitters, outside the cell.

Organelles and the nucleus

- Ribosomes make proteins; free ribosomes make proteins for the cytosol, while ribosomes on rough ER make proteins for secretion or membranes.
- Smooth ER makes lipids and steroids and detoxifies; the Golgi apparatus modifies, sorts, and packages proteins.
- Mitochondria make most ATP by aerobic respiration; lysosomes digest with acidic enzymes; peroxisomes break down fatty acids and hydrogen peroxide.
- Cytoskeleton: microfilaments (actin, thinnest), intermediate filaments (tension), and microtubules (thickest; spindle, cilia, flagella).
- The nucleus holds DNA as chromatin inside a double nuclear envelope with pores; the nucleolus builds ribosomal subunits.

From DNA to protein

- DNA replication occurs in S phase and is semiconservative; DNA polymerase adds complementary nucleotides.
- Transcription in the nucleus: RNA polymerase makes mRNA from a DNA template; introns are spliced out and exons kept.
- Translation at the ribosome: tRNA anticodons pair with mRNA codons and deliver amino acids.
- A codon is three nucleotides; AUG is the start codon (methionine); UAA, UAG, and UGA are stop codons.
- The code is redundant: 64 codons specify 20 amino acids, so several codons can code for one amino acid.

Cell cycle, mitosis, and differentiation

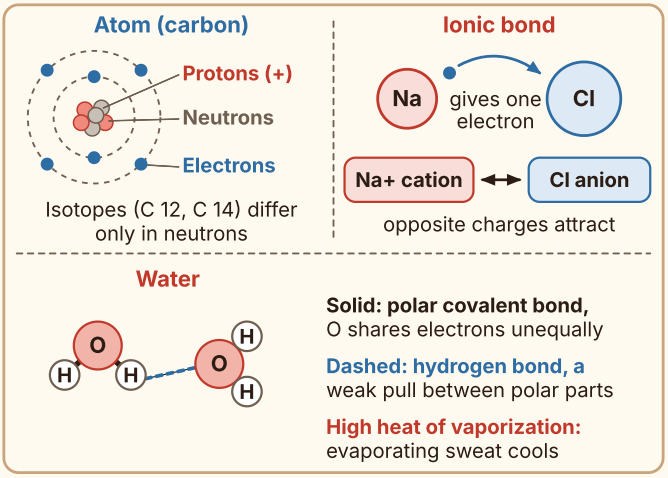
- Interphase: G1 (growth), S (DNA synthesis), G2 (preparation); cells that stop dividing enter G0.
- Mitosis phases: prophase (chromosomes condense, spindle forms), metaphase (chromosomes line up at the equator), anaphase (sister chromatids separate), telophase (nuclear envelopes reform).
- Cytokinesis divides the cytoplasm using a contractile ring of actin that forms a cleavage furrow.
- Mitosis produces two genetically identical daughter cells with 46 chromosomes each.
- Stem cells: totipotent (zygote, all cells plus placenta), pluripotent (all body cells), multipotent (one family, such as blood cells), unipotent (one cell type).
- Differentiation results from different patterns of gene expression, not from different DNA.

Villager wisdom

- Old Mira the potion maker says an atom that gives away an electron turns positive, and one that takes an electron turns negative. Opposites cling together in ionic bonds.
- Remember the pH scale runs backward: the more hydrogen ions in the brew, the lower the number and the more acidic it is. Healthy blood stays between 7.35 and 7.45.
- Water moves by osmosis toward the side with more dissolved solute. Drop a cell in a hypotonic pond and it swells; put it in a hypertonic sea and it shrivels.
- A cell that ships out proteins is like our mill: rough ER builds the goods, and the Golgi packs and labels them for delivery.
- DNA is copied in S phase, written into mRNA in the nucleus, and read into protein at the ribosome, where tRNA anticodons match each codon.

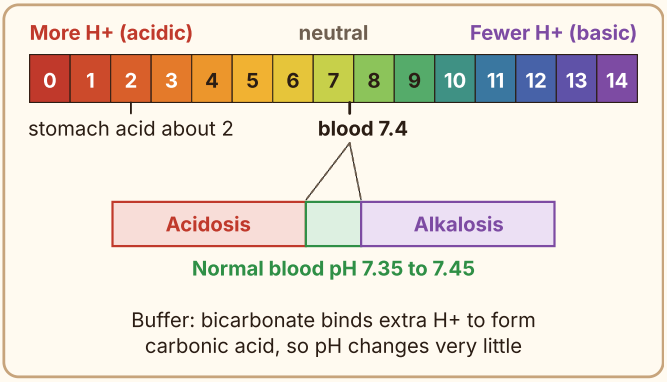
Diagrams

Atoms, ions and chemical bonds



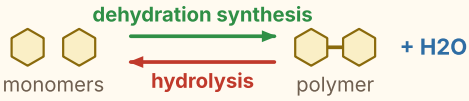
Atoms of an element share a proton number (isotopes differ in neutrons); losing an electron makes a cation, ionic bonds join opposite charges, and polar water molecules link by hydrogen bonds.

The pH scale



Each pH unit is a tenfold change in H+; blood stays between 7.35 and 7.45, so a pH of 7.25 means extra H+ and acidosis.

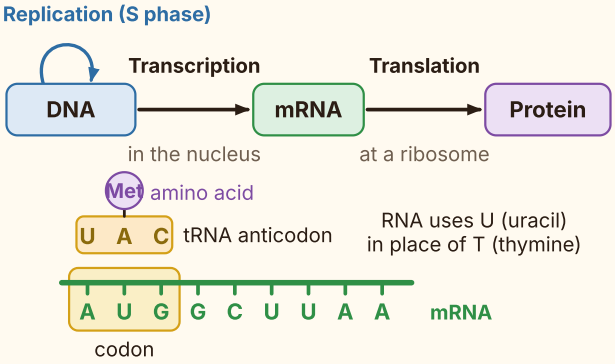
Building blocks of macromolecules



Carbohydrates	Monosaccharides glucose, fructose (simple sugars)	Sucrose = glucose plus fructose; glycogen stores glucose
Lipids	Glycerol plus three fatty acids make a triglyceride	Saturated: no double bonds, solid (butter); steroids: cholesterol
Proteins	Amino acids joined by peptide bonds	Amino acid sequence = primary structure; denaturing unfolds it
Nucleic acids	Nucleotides: sugar, phosphate and a nitrogen base	DNA and RNA store and express genetic information

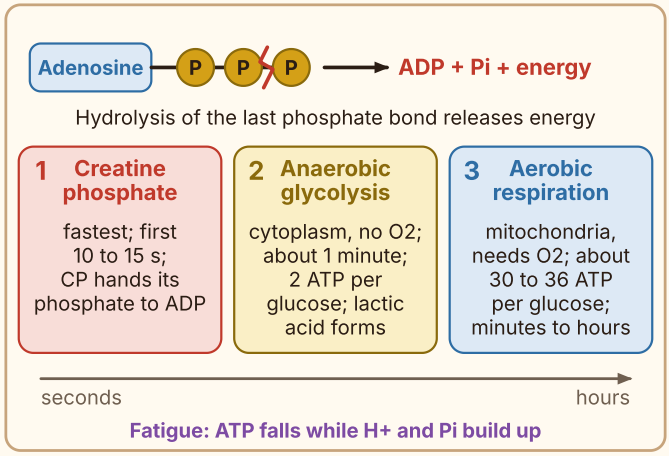
Monomers join by dehydration synthesis (releasing water) and polymers split by hydrolysis (adding water); each class of macromolecule has its own building block.

DNA to RNA to protein



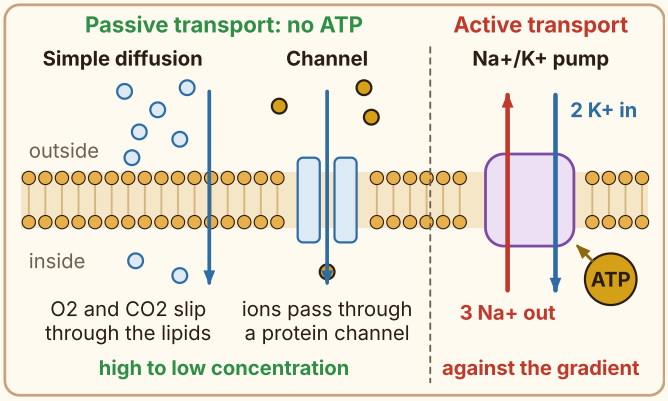
Transcription copies a gene into mRNA in the nucleus; at the ribosome each mRNA codon pairs with a tRNA anticodon that brings the matching amino acid.

ATP and muscle energy systems



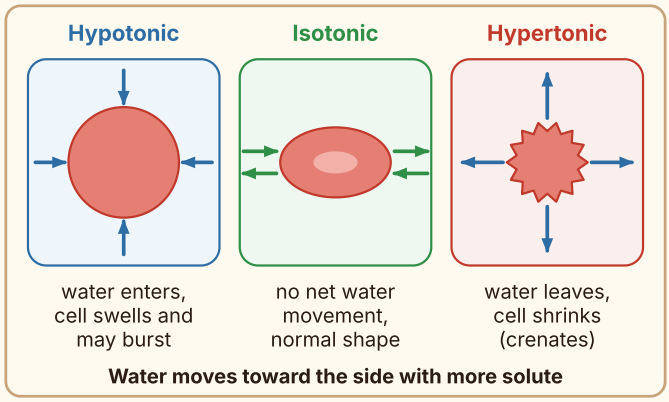
Muscle first rebuilds ATP from creatine phosphate, then from anaerobic glycolysis for about a minute, and for longer activity from aerobic respiration in the mitochondria, which yields far more ATP.

Passive and active transport



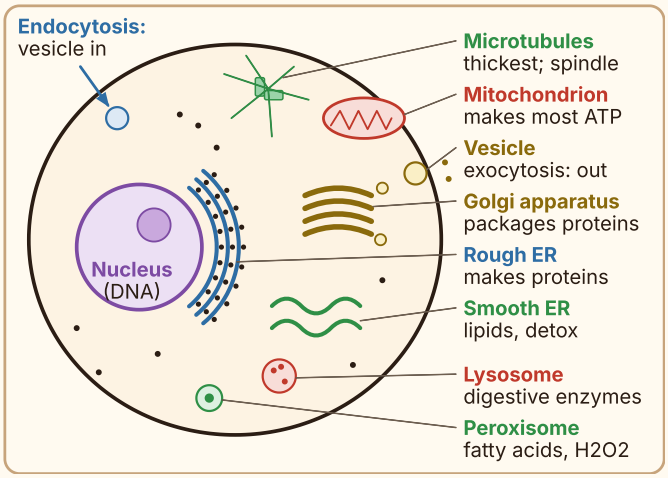
Passive transport moves substances down their concentration gradient with no ATP; the sodium potassium pump uses ATP to move 3 Na+ out and 2 K+ in against their gradients.

Osmosis and tonicity



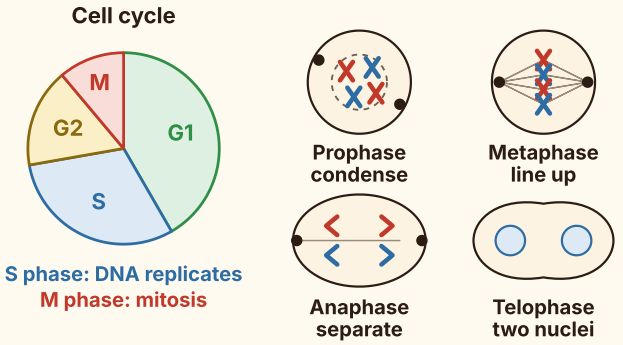
Water moves by osmosis toward more solute: in a hypotonic solution cells gain water and swell, in an isotonic solution nothing changes, and in a hypertonic solution cells shrink.

Animal cell organelles



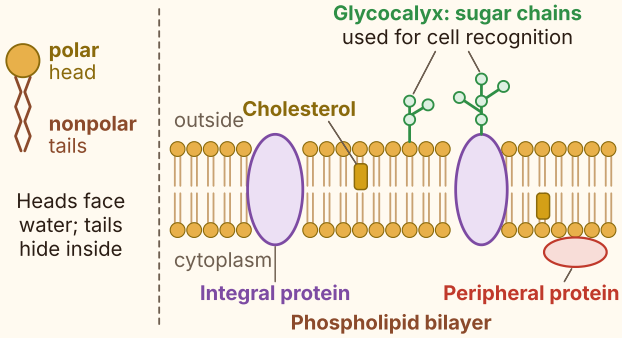
The nucleus holds DNA; rough ER and the Golgi make and ship proteins, smooth ER makes lipids, mitochondria make most ATP, and lysosomes digest with enzymes.

Cell cycle and mitosis



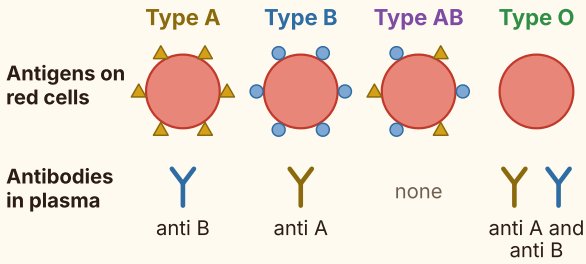
DNA is copied in S phase of interphase; in mitosis chromosomes condense (prophase), line up at the center (metaphase), separate (anaphase), and form two nuclei (telophase).

The plasma membrane



The fluid mosaic model: a phospholipid bilayer with proteins floating in it; sugar chains on the outer surface form the glycocalyx, which the body uses for cell recognition.

ABO and Rh blood groups

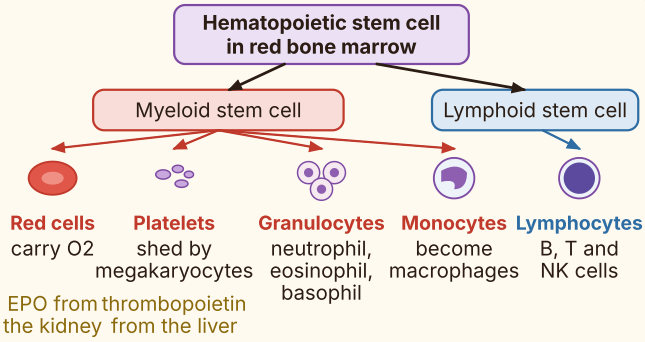


Rh: an Rh negative mother exposed to Rh positive blood makes anti D. Anti D IgG can cross the placenta and attack red cells of a later Rh positive fetus.

RhoGAM given to the mother prevents this.

Plasma holds antibodies against the ABO antigens a person lacks, so type A blood carries anti B and attacks type B donor cells.

Hematopoiesis



All formed elements come from hematopoietic stem cells in red bone marrow; erythropoietin from the kidney drives red cell production and thrombopoietin from the liver drives platelet production.

Self check

1. **Which region inside the nucleus makes ribosomal RNA and assembles ribosomal subunits?**
    A. Nuclear envelope
    B. Nucleolus
    C. Nuclear pore
    D. Centrosome
2. **Why do phospholipids arrange themselves into a bilayer in water?**
    A. Nonpolar heads face water; polar tails hide inside
    B. Both heads and tails are strongly hydrophilic
    C. They bond covalently to surrounding water
    D. Polar heads face water; nonpolar tails hide inside
3. **In Tay Sachs disease, lipids build up in neurons because one digestive enzyme is missing. Which organelle is malfunctioning?**
    A. Peroxisome
    B. Mitochondrion
    C. Golgi apparatus
    D. Lysosome
4. **A patient's diet is high in fats that are solid at room temperature, such as butter and lard. These fats contain mostly:**
    A. Unsaturated fatty acids with double bonds
    B. Phospholipids with polar heads
    C. Saturated fatty acids with no double bonds
    D. Steroids built from four rings
5. **The fluid mosaic model describes the plasma membrane as:**
    A. A phospholipid bilayer with embedded proteins
    B. A rigid sheet made only of protein
    C. A single layer of cholesterol
    D. A double layer of carbohydrates
6. **A solution at pH 4 has how many times more hydrogen ions than a solution at pH 6?**
    A. 100 times
    B. 2 times
    C. 10 times
    D. 1,000 times
7. **Which cytoskeletal element is the thickest and forms the mitotic spindle?**
    A. Microfilaments
    B. Intermediate filaments
    C. Ribosomes
    D. Microtubules

8. **During translation, what pairs with an mRNA codon at the ribosome?**
    A. A DNA promoter
    B. A ribosomal protein
    C. A tRNA anticodon
    D. An RNA polymerase
9. **Enzymes speed up chemical reactions by:**
    A. Raising the temperature of the cell
    B. Being used up in the reaction
    C. Lowering the activation energy
    D. Supplying ATP to the reactants
10. **An ionic bond forms when:**
    A. Two atoms share electrons equally
    B. Two atoms share electrons unequally
    C. Hydrogen is drawn to oxygen of another molecule
    D. Oppositely charged ions attract each other
11. **The primary structure of a protein is its:**
    A. Alpha helices and pleated sheets
    B. Overall three dimensional folded shape
    C. Sequence of amino acids
    D. Grouping of several polypeptide chains
12. **In human cells, where does transcription of DNA into mRNA take place?**
    A. Nucleus
    B. Ribosome
    C. Golgi apparatus
    D. Lysosome

Answer key

1. **B.** Nucleolus. The nucleolus is a dense region of the nucleus that builds ribosomal subunits. Nuclear pores only let molecules pass in and out; they do not build ribosomes.
2. **D.** Polar heads face water; nonpolar tails hide inside. The phosphate head is hydrophilic and the fatty acid tails are hydrophobic, so the tails cluster away from water between two layers of heads. The reversed arrangement has the polarity backward.
3. **D.** Lysosome. Lysosomal enzymes normally break down specific lipids; when one is missing, undigested lipid accumulates and damages neurons. Peroxisomes break down fatty acids but are not the site of this enzyme.
4. **C.** Saturated fatty acids with no double bonds. Saturated fatty acids have only single bonds, so their straight chains pack tightly and stay solid at room temperature. Unsaturated fats have double bonds that kink the chains, which keeps oils liquid.
5. **A.** A phospholipid bilayer with embedded proteins. The membrane is a fluid phospholipid bilayer in which proteins, cholesterol, and attached carbohydrates move and float like tiles in a mosaic. Proteins are part of the mosaic, but the membrane is not made of protein alone.
6. **A.** 100 times. The pH scale is logarithmic, so each whole unit is a tenfold change; two units means 10 times 10, or 100 times. Ten times would be correct only for a difference of one pH unit.

7. **D.** Microtubules. Microtubules are hollow tubes of tubulin that form the spindle, cilia, and flagella. Microfilaments are the thinnest elements, made of actin, and work in contraction and cytokinesis.
8. **C.** A tRNA anticodon. Each tRNA carries one amino acid and has a three base anticodon that pairs with the matching mRNA codon. RNA polymerase works in transcription, not translation.
9. **C.** Lowering the activation energy. Enzymes position substrates so less energy is needed to start the reaction, which makes it go faster. They are not consumed, so the same enzyme can catalyze the reaction again and again.
10. **D.** Oppositely charged ions attract each other. An ionic bond is the attraction between a cation and an anion after electrons are transferred, as in sodium chloride. Sharing electrons, equally or unequally, describes covalent bonds.
11. **C.** Sequence of amino acids. Primary structure is the order of amino acids linked by peptide bonds, which determines all higher folding. Helices and pleated sheets are secondary structure, formed by hydrogen bonds along the chain.
12. **A.** Nucleus. DNA stays in the nucleus, so RNA polymerase copies genes into mRNA there. The ribosome is where the next step, translation, happens after mRNA leaves the nucleus.

Kingdom of Corpus · Dr. Akash Garg, Dr. Garg Learning · Aligned to OpenStax Anatomy and Physiology 2e · CC BY SA 4.0